

Augmented Fishing Technique for Reducing Overfishing

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1 INTRODUCTION

Augmented reality (AR) provides the opportunity for users to interact with computer-generated content from the real world. The application of AR technology can be seen in a variety of areas, such as news, travel, sports, and commercials, in which tangible and exciting experiences are created [3].

The development of smart fishing rods attracted many fishing equipment manufacturers, many companies produce smart fish finders, while the development of smart fishing equipment are certainly not new, their increasingly widespread deployment does not emphasize the need to consider the fishing wealth, overfishing, and rules and regulations.

Ecological extinction caused by overfishing precedes all other pervasive [6], can we utilize technology to reduce overfishing? Is inadvertent bad practices related to unawareness and lack of ongoing information one of the main reasons for overfishing that we can reduce? While Fishing Sport is highly regulated, fishermen need a real-life automation and guidance for their fishing practices.

Traditional guidance in Fishing Sport is showing a limited approach to raise awareness for fishermen professionals, practicing fishing professionally taking into consideration legal issues and nature preservation is putting pressure on fishermen to be aware of many details like fish type, age, season of fishing, areas of fishing with the variety of fish types and unlimited number of lakes, seas and oceans makes the task complicated.

Through a survey of current fishermen professionals, we identify a significant need to improve ways of fishing to allow interactive exploration of underwater environment, fishermen reported their inability to avoid inadvertent practices due to lack of ongoing and on spot information, also the fishermen connection to the underwater is diminished.

In this research study we introduce augmented fishing as a tool to reduce overfishing, and we discuss how Augmented Reality can help in reducing problems related to fishing wealth, and how it can allow fishermen abide with fishing rules and regulations related to fishing season, size, age, and area of fishing.

Two studies were conducted using two groups of fishing professionals and fishing practitioners; our first study is conducted using traditional fishing aiming to highlight and record inadvertent bad practices committed by the groups, where the fishermen have limited information and normal information they get from signs and official webpages, and we compare it with the second study which we conducted using Augmented Fishing where the groups are exploring ongoing information, alarms, and textual augmentation for location and underwater activities through their fishing practices.

This work addresses the inadvertent fishing practices that cause overfishing reported by fishermen, with a novel approach to partially blending augmented reality with traditional fishing to reduce

and eliminate such practices. We present Augmented Fishing as a tool to reduce these practices that cause overfishing problems.

2 RELATED WORK

Prior work has examined to show the highlighted fishing related problems that impact on many aspects such as, the fishing pressure and the overfishing problems. The authors in [1, 8, 10] focused on the study of illegal, unreported and unregulated (IUU) fishing activities that became a high priority in the international fisheries management agenda, and have a clear academic contributions to improve the understanding of the problem. Such that, the total value of current illegal and unreported fishing losses worldwide is between 10 billion dollars and 23.5 billion dollars annually. In [8], the authors focused on the analytical framework developed regarding the causes of and the potential solutions to this widespread problem. They have recognized that this framework wasn't fitted to solve the problem completely, but alternative measures must be worked out, including the establishment of cooperation and development programs. While the authors in [10] concentrated their efforts in proposing multiple quantification methods that are based on information on gear use and spatial distribution. They claimed that integrating fisher's knowledge and involving fishers in data collection and management decisions may be the most effective way to improve data quality and accessibility. And finally the authors in [1] addressed the need for international fishing agreements between underdeveloped and developed countries to raise awareness and formulate strategies to curb illegal fishing activities.

Moreover, regarding the data analysis, the authors in [12] depended on size-based indicators (SBIs) to indicate the heavy fishing pressure, but they concluded that no single SBI can serve as an effective overall indicator of heavy fishing pressure. Rather, suites of SBI should be selected. This factor is very dependent on the gear used, time, and location, which restricts their interpretation to consistently collected time-series data.

In the other direction, the authors in [11] attempted to find the criteria behind fishing abundance, they applied the classical catch equation to fisheries data for the purpose of measuring relative abundance or density changes that ended up with an unexplained variability. So that they proposed reformulation and study of catch data in terms which explicitly recognize the heterogeneity in fish distributions and directed fishing operations. In it, they concluded that this fishing and its related problems behavior is most related with that fishes are randomly distribution in such schools on a population area. Fishermen, with a given radius of perception, search for these schools, detecting them with given probability, and upon successful detection, fish out a certain proportion, and the other fishermen without an enough fishing experience or perception are made this in a random way as well which lead to unintended problems as overfishing.

Furthermore, the large variation on fishing behavior has been detected as an area based behavior, so other authors decided to be more specific in their studies to be area focused. Such that in [2], the authors analyzed a unique dataset, combining observer data for Irish, French and British vessels operating in the Celtic Sea between 2010 and 2015. They implemented a mobile application with an incorporated maps to identify where catches of quota restricted species to better inform decision making and potentially increase fishing opportunities under the Landing Obligation. This optimised the fishing practices in terms of time and location fished. This solution depends on fishermen awareness and their continuous checking for this map while practicing, but sometimes the results were not accurate and misleading. For this reason, the need of real-time solutions are required in this direction. Following the suggested approach, the authors in [13] analyzed observer data from the Spanish discard sampling program from 2004 to 2008 for several species caught in the Cantabrian Sea, but this time presented near-real-time maps. Such that, a statistical tool was introduced, based on the random forest technique, that aims to provide fishermen with near-real-time maps of a fishing suitability index based on haul-by-haul catch and discard rates, indicating the most suitable areas for fishing.

The urgent need for a real-time underwater visualization was most fitted in the virtual and augmented reality contribution area. For this reason, the journey was started with fishing simulations to the way to the reality augmentation.

A group of researchers located in Taiwan implemented a virtual fishing system [5] that helped people to virtually practice fishing through virtual reality, they presented an interested virtual fishing system in the paper, which a low-cost 3-D parallel platform is used to simulate the fishing boat and the force interaction is provided by a PMSM and its drive through a fishing rod, the gesture of the platform is adjusted by fuzzy control, and the virtual scenes are animated by VR software packages.

The authors in [4] designed and implemented a full featured virtual fishing system in an effort to simulate real fishing experience to make people enjoy the fun of fishing, which is also a brand new attempt to integrate embedded technology and virtual technology.

Continuing in such enrichment for the fishing virtually, the authors in [7] reported the development of a Haptic device of virtual fishing system for entertainment and imitated real fish motion and to feel the fishing force. they also incorporated a stereo vision to detect the end position of a fishing rod. Following the same approach of Haptic interface for fishing training, the authors in [Lin] developed a haptic interface based simulation for force sensation during fishing. It simulated the dynamic objects such as fishing rod, hook and fish. Furthermore, this interface helped beginners got more practice before they turn to real fishing.

To the best of our knowledge, there are no existing researches or implementations for augmented fishing through annotated textual for classified fish images and current fishing location. For this reason, it is the time for our proposed methodology to merge the real fishing with a technological solution based on augmentation to produce an augmented fishing real-time solution.

3 METHOD SECTION

We developed an augmented based methodology to reduce the overfishing problem and to cover the fishing usability concerns. Moreover, a stage of the outcome investigation to measure the impact of this proposed technique on different professionalism levels.

3.1 Participants

Thirty paid participants participated in the study. They were divided into two groups, the first one consisting of twenty professional fishermen with 5 - 10 years of experience in fishing, and the other one had ten fishing practitioners with 10 - 15 previous fishing activities. Their average age was 30 years with standard deviation of 10.5. In addition to a control group consisted on five men from the regularity parties were recruited to monitor and control the procedure correctness and validation.

All participants were living in North America, and they were aware of the fishing rules and regulations in their areas. They applied through a public advertisement posted in fishing related groups in social media. Their ability to use smart phones and smart devices appropriately is the only precondition besides their required prior fishing experience.

3.2 Materials

The study was conducted with a fishing rod of a fishing string 1000 meters long, which had a camera attached to it. A Nano SeaCam was used with the following specifications: 2.54 mm Fixed Focus, F/3.5, 620 TVL resolution, Up to 11,000 m depth rated, and 100 Foot Transmit Range via WiFi. Such that, this camera was linked via WiFi to an installed application on a smartphone or a smart monitor, which was transmitting the underwater stream to a cloud based application for the required recognition purpose.

We wrote this application to employ the Machine Learning (ML) classification algorithms to analyze and recognize specific attributes from the recorded video stream. Furthermore, the fish classification attributes contained Name, Species, Family, Size, Age, Season, and origin of fishes, that were considered to construct the suitable warnings and annotations against the established fishing regulations. These conclusions were sent back to the smart application to describe the current underwater state to the participants. For example, Figure 1 shows these different parts connected together, and Figure 2 to show how the annotations and states were described through the smart monitor.

3.3 Design

The experiment used a repeated measures design (within-subject design). It considered the usage of the augmented fishing technique as an independent variable in addition to the level of experience for the participants affected the overfishing rate dependent variable. Such that the professional participants were expected to be aware of fishing regulations to avoid such overfishing problems more than the other group of fishing practitioners.

Moreover, the fishing area and the fishing time were considered as the control variables to be held constant while comparing the overfishing rate result. The variation on the independent variables affected the overfishing rate in either direction.

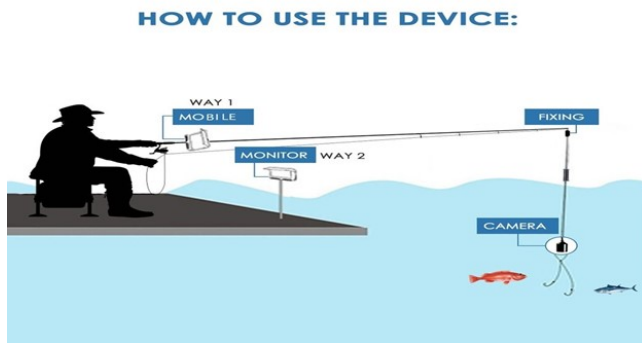


Figure 1: Augmented Fishing Prototype.



Figure 2: Prohibited fishing state through a smart monitor.

The experiment showed that the usage of augmented fishing technique increased the professionalism level of a fisherman, and eventually decreasing the inadvertent overfishing rate. Fishing area and time were considered constant during the experiment.

3.4 Procedure

The procedure was conducted through two different groups of participants, G1 had the professional fishermen, and G2 included the fishing practitioners as described above in the participants section.

The fishing area was considered the same per experiment which was Outer Banks, North Carolina, in addition to the fishing time scheduled to be in the morning. Each experiment lasted 1.5 hours long to be conducted.

The overfishing rate was calculated through the number of wrong fishing for prohibited fishes divided by the total number of fished fishes during the experiment. For this reason, a control group of five men from the regularity parties were recruited to monitor and ensure the wrong and correctness of such fishing trials.

The procedure included two different studies each one measured the influence of specific variable to the overfishing rate as following:

Study 1

In this study we focused on how the professionalism level influenced the overfishing rate, such that G1 and G2 were asked to start their fishing activity based on the above conditions. This fishing session

was recorded using a video camera. Moreover, the number of wrong or prohibited fishing trials, in addition to the number of valid and correct fishing trails were measured per group. After that, an overall overfishing rate was conducted per group.

This study showed that G1 reached an overfishing rate with 30%, while G2 reached a higher rate with 65% during this session. It was sound as an expected variation between these groups, because G1 had much experience in fishing and much awareness for fishing regulations as well. In contrast, G2 didn't practice fishing in a frequent way to achieve it in a better way. After the end of this experiment, we conducted a survey with both fishermen groups to identify the required from their perspectives to reduce these overfishing rates. Through their suggestions, we identified a significant need to improve ways of fishing to allow interactive exploration of the underwater environment, fishermen reported their inability to avoid inadvertent practices due to lack of on spot information, also the fishermen connection to the underwater was diminished.

Study 2

This study introduced the augmented fishing technique and materials to be used by the fishermen. G1 and G2 were directed to check the mentioned apparatus, such that each individual fisherman was allowed to try it freely for 30 minutes without any instructions. After that, one hour session was held to answer the most relevant questions and inquiries that may affect their fishing experience during the experiment.

There were two major concerns, the first one was a usability issue which was related to attaching the monitor to the fishing rod regardless of the variation of its size and weight as shown in figure 3. For solving this one we added the ability to place the monitor on a solid surface like the ground besides the fisherman. The second concern was related to WiFi coverage and speed limitations, so we provided a fixed hotspot for this fishing area to ensure the network stability during the experiment.



Figure 3: Attaching a monitor device to the fishing rod.

Each group started this fishing experiment with the matched conditions as same as the first study, but this time using the augmented fishing technique. The recording and monitoring for the overfishing rate were performed during the fishing session.

3.5 Results

The results of these studies showed how the proposed augmented fishing methodology positively impacted on the overfishing rate, such that G1 overfishing rate decreased with the half of its previous value to be 15% instead of 30%, while the surprise was derived from G2 which reached to 20% to be nearly the same as the professional group rate, figure 4 illustrated such comparison.

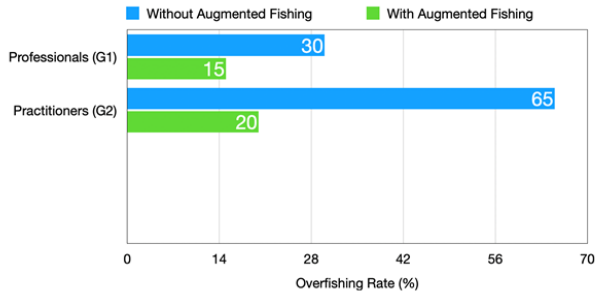


Figure 4: How overfishing rate influences through both studies

In conclusion, the usage of augmented fishing had a great positive effect on decreasing the overfishing rate, in addition to making the fishermen perform the fishing in a professional and a controlled way regardless the different level of experience among them.

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